(to prepare for Midterm #1 on Thursday, Sep. 29, 2016

1: Done in class on Feb. 14, 2017.

2: Done in class on Feb. 14, 2017.

3: $p := 329867$ turns out to be prime.

(a) Please verify that 2 is a generator for $\mathbb{F}_p^*$, by factoring $p-1$ and checking just a few powers of 2 mod p (using the technique alluded to in Problem 4b from HW#3).

Via Maple (or simply dividing by primes up to 71) one easily sees that $p-1 = 2 \cdot 23 \cdot 71 \cdot 101$. So then, we merely need to check the values of $2^{\frac{329866}{2}}$, $2^{\frac{329866}{23}}$, $2^{\frac{329866}{71}}$, and $2^{\frac{329866}{101}}$ mod 329867 (following the trick mentioned in homework and in class several times now). These values turn out to be, respectively, $2^{164933} = -1$, $2^{14342} = 3966$, $2^{4646} = 224774$, and $2^{3266} = 212199$ mod 329867. So it is impossible for the order of 2 in $\mathbb{F}_{329867}^*$ to be smaller than 329866, and thus 2 is a generator. Note also that $164933 < 2^{18}$, so computing the preceding three powers can be done via 17 squarings mod 329867, along with no more than $4 \cdot 16 = 64$ additional multiplications mod 329867. (The actual number of multiplies depends on the number of 1s in the binary expansions of 164933, 14342, 4646, and 3266: The total is $6+5+5+5=21$, so the number of extra multiplies is in fact $5+4+4+4=17$.)

(b) Please solve for the unique $x \in \{2, \ldots, 329866\}$ satisfying $2^x = 11$ mod 329867 via the Pohlig-Hellman Method.

Knowing that $p-1 = 2 \cdot 23 \cdot 71 \cdot 101$ means we can reduce our DLP over $\mathbb{F}_p^*$ to 4 separate smaller DLPS. These smaller DLPs will be over subgroups of respective sizes 2, 23, 71, and 101. In particular, to set up these smaller problems, we need to take both sides of the original equation $2^x = 11$ to the following powers: $\frac{329866}{2} = 164933$, $\frac{329866}{23} = 14342$, $\frac{329866}{71} = 4646$, and $\frac{329866}{101} = 3266$. We then obtain the following smaller DLPs:

$$(2^{164933})^{x_1} = 11^{164933} \text{ mod } 329867$$
$$(2^{14342})^{x_2} = 11^{14342} \text{ mod } 329867$$
$$(2^{4646})^{x_3} = 11^{4646} \text{ mod } 329867$$
$$(2^{3266})^{x_4} = 11^{3266} \text{ mod } 329867$$

More explicitly, evaluating the 8 powers above, our 4 DLPs are:

$$329866^{x_1} = 1 \text{ mod } 329867$$
$$3966^{x_2} = 85713 \text{ mod } 329867$$
$$224774^{x_3} = 155982 \text{ mod } 329867$$
$$212199^{x_4} = 254273 \text{ mod } 329867$$

Of course, one may wonder how exactly these 4 new DLPs are any smaller: They are smaller because of the allowable values of the x_i. In particular, $x_1 \in \{0, 1\}$, $x_2 \in \{0, \ldots, 22\}$, $x_3 \in \{0, \ldots, 70\}$, and $x_4 \in \{0, \ldots, 100\}$. (We thus need only search over $2+2+2+2+1+22+70+100 = 203$ powers instead of 329866 powers!) So now, we need to find the appropriate values of x_1, x_2, x_3, x_4.

This can be done either by brute-force or Baby-Steps/Giant-Steps. To keep things simple, I will simply state the appropriate values (which I found via a brute-force search using Maple): $x_1 = 0$, $x_2 = 21$, $x_3 = 43$, $x_4 = 79$.

At this stage, we apply the Chinese Remainder Theorem to find the unique $x \in \{0, \ldots, 329866\}$ with

$$x = 0 \bmod 2 \quad , \quad x = 21 \bmod 23 \quad , \quad x = 43 \bmod 71 \quad , \quad x = 79 \bmod 101$$

This simply means that we compute

$$x = 0 \cdot (23 \cdot 71 \cdot 101)\left(\tfrac{1}{23\cdot 71\cdot 101} \bmod 2\right) + 21 \cdot (2 \cdot 71 \cdot 101)\left(\tfrac{1}{2\cdot 71\cdot 101} \bmod 23\right) + 43 \cdot (2 \cdot 23 \cdot 101)\left(\tfrac{1}{2\cdot 23\cdot 101} \bmod 71\right) + 79 \cdot (2 \cdot 23 \cdot 71)\left(\tfrac{1}{2\cdot 23\cdot 71} \bmod 101\right)$$

$= 16580744 = 87444 \bmod 329866$. So our final answer is $\boxed{x = 87444}$ and we can in fact check via Maple that $2^{87444} = 11 \bmod 329867$.

Note: On a midterm, the numbers would be nowhere this big: You would be able to do any necessary computations by hand. This example was specifically chosen to be large so as to illustrate the underlying ideas in greater detail.

4: Suppose $m_1, \ldots, m_k \in \mathbb{N}$ are pair-wise relatively prime, and $a_1, \ldots, a_k \in \mathbb{Z}$. Please prove that if $x_1, x_2 \in \mathbb{Z}$ are solutions to the simultaneous congruences $x = a_1 \bmod m_1, \ldots, x = a_k \bmod m_k$, then $x_1 = x_2 \bmod \prod_{i=1}^k m_i$.

If we have $x_1 = a_i = x_2 \bmod m_i$ for all i, then we clearly have $x_1 = x_2 \bmod m_i$ for all i. In particular, this means that $x_1 - x_2 = 0 \bmod i$ for all i. Put another way, this means that $x_1 - x_2$ is divisible by m_i for all i. Since the m_i are pair-wise relatively prime, this means that $x_1 - x_2$ is divisible by $\prod_{i=1}^k m_i$. In other words, $x_1 = x_2 \bmod \prod_{i=1}^k m_i$.

5: Please prove Euler's Theorem: If $a, n \in \mathbb{N}$ with $\gcd(a, n) = 1$ then $a^{\varphi(n)} = 1 \bmod n$.

Recall that $(\mathbb{Z}/n\mathbb{Z})^*$ is the multiplicative group of integers in $\{1, \ldots, n-1\}$ relatively prime to n, and that $\varphi(n)$ is the cardinality of $(\mathbb{Z}/n\mathbb{Z})^*$. In particular, let us denote the $\varphi(n)$ elements of $(\mathbb{Z}/n\mathbb{Z})^*$ by $\alpha_1, \ldots, \alpha_{\varphi(n)}$. Now consider the sets $(\mathbb{Z}/n\mathbb{Z})^*$ and $S := \{a\alpha_1, \ldots, a\alpha_{\varphi(n)}\}$: The key to our proof will be showing that these two sets are identical. In particular, first note that $S = f((\mathbb{Z}/n\mathbb{Z})^*)$, where $f(x) := ax \bmod n$. So we'd like to show that f is a bijection.

To see that f is injective, note that $f(x_1) = f(x_2)$ implies $ax_1 = ax_2 \bmod n$ and thus $x_1 = x_2 \bmod n$ since a is invertible mod n.

To see that f is onto, simply note that $f(x) = y$ implies that $ax = y \bmod n$, which in turn implies that $x = y/a \bmod n$, and the latter formula for x is well-defined since a is invertible mod n.

Knowing that f is a bijection, we thus obtain $S = (\mathbb{Z}/n\mathbb{Z})^*$ as sets, i.e., the sequences $a\alpha_1, \ldots, a\alpha_{\varphi(n)}$ and $\alpha_1, \ldots, \alpha_{\varphi(n)}$ are the same, up to re-ordering. To conclude, consider the product $P := (a\alpha_1) \cdots (a\alpha_{\varphi(n)}) \bmod n$. On the one hand, since $S = (\mathbb{Z}/n\mathbb{Z})^*$, the product P is nothing more than $\alpha_1 \cdots \alpha_{\varphi(n)}$. On the other hand, commutativity tells us that $P = a^{\varphi(n)}(\alpha_1 \cdots \alpha_n)$. So we obtain $a^{\varphi(n)}(\alpha_1 \cdots \alpha_n) = (\alpha_1 \cdots \alpha_n) \bmod n$. Since each α_i is invertible mod n, we can divide by $\alpha_1 \cdots \alpha_n$ to finally obtain $a^{\varphi(n)} = 1 \bmod n$.

6: (a) Pierre Dusart's 1998 Ph.D. thesis from the university of Limoges provides (among many other results) the following lower and upper bounds on the prime-counting function $\pi(x)$: $\frac{x}{\log x}\left(1 + \frac{1}{\log x}\right) < \pi(x) < \frac{x}{\log x}\left(1 + \frac{1}{\log x} + \frac{2.51}{(\log x)^2}\right)$, for all $x \geq 355991$. Please use this estimate to find lower and upper bounds for the number of 100-digit primes.

By the definition of the prime counting function, we simply need to estimate $\pi(10^{101} - 1) - \pi(10^{100} - 1)$. Let us recall the following basic fact about inequalities: If we are estimating a quantity $a - b$, with $A_- \leq a \leq A_+$ and $B_- \leq b \leq B_+$, then

clearly we have $A_- - B_+ \leq a - b \leq A_+ - B_-$.
Applying Dusart's formula via Maple (with precision set to 101 digits), we then get that $\pi(10^{101}-1)$ lies in the interval
$[4318434895523972994998919874575254797117929431560292487648440444990041655888531639879709572818 45221, 4318633655447613154482422935514841340555294469455995220511757613831394706003592379575438136041855 30]$,
while $\pi(10^{100}-1)$ lies in the interval
$[4361805988733679669433285272162556910310560112554287710038728719617755028251004048368369240298 2600, 4362010771281766049389297959351673865600620748463268131410537261992169917137601243537618074715895 3]$.
So then, the number of 100-digit primes is an integer in the interval
$\boxed{[3882233818395796390059990078640087410557867356713965674507386718790824664174771515525947765346 86268, 38824530565742451875390944082985856495242384582005664495078847418696192031784919747386012120120 2930]}$.
Note in particular that while the leading 4 digits of the upper and lower bounds agree, the discrepancy is a 95-digit integer.

(b) The (arguably) most important open problem in mathematics is the Riemann Hypothesis. Among many other implications, if true, it would imply the following sharper estimate for $\pi(x)$: $\quad \mathrm{li}(x) - \frac{\sqrt{x}\log x}{8\pi} < \pi(x) < \mathrm{li}(x) + \frac{\sqrt{x}\log x}{8\pi}, \quad$ for all $x \geq 2657$, where $\mathrm{li}(x)$ is a well-known integral that can be evaluated easily within `Maple`. (This conditional estimate was published by Schoenfeld in 1976.) In particular, the commands:

```
Digits:=100;
evalf(Li(10^100));
```

will evaluate $\mathrm{li}(10^{100})$ to 100 digits of accuracy. Please use the preceding (conditional) estimate to find lower and upper bounds for the number of 100-digit primes. Are your estimates significantly sharper than those from Part (a)?
Applying Maple in the same way as Part (b), one runs into a small bug: Maple 14 leaves floor(Li(10^{101}-1)) (and a few other similar quantities) unevaluated. However, if one applies the evalf() command, then one gets a floating point answer. Proceeding this way, we obtain that the number of 100-digit primes is (hopefully) an integer in the interval
$\boxed{[3882398791983634761451408174842585354066173386875152288932042413339121100547879929017168531141 10624, 3882398791983634761451408174842585354066173386951998762138661408134307029533880108162726778395 14934]}$.
Note in particular that for these (GRH-dependent) upper and lower bounds, <u>46</u> of their leading digits agree. So now the discrepancy is a 52-digit integer, and thus dozens of orders of magnitude better than Part (a). Note, however, that Maple's computation of Li() and log() should be verified further if we really want to believe this estimate. In fact, more generally, one should always verify one's computations if the answer is of great importance.